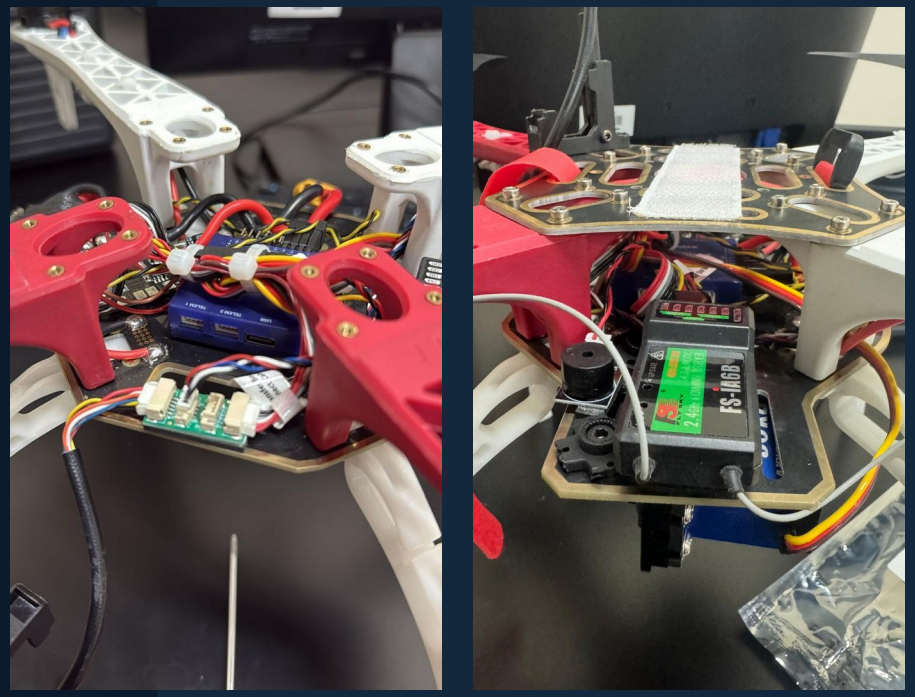
PROJECT PROGRESS | 09 NOVEMBER 2026

SILENT WILDLIFE UAV

Low-noise propulsion + onboard animal detection

Mansour Alnaqbi · Shamma Nasser · Alyazia Alshehhi · Salim Alkathiri

Supervisor and course professor: Dr. Claudio Vignola



Current quadcopter | assembly photos

PROJECT PURPOSE

Track wildlife with less disturbance

Our design prioritizes acoustic performance, reliable detection and usable flight endurance.

01

QUIETER FLIGHT

Investigate toroidal propellers to reduce acoustic disturbance near animals.

02

ONBOARD DETECTION

Use YOLO to recognize animals locally without continuous video transmission.

03

PLANNED SURVEYS

Follow a Mission Planner waypoint route using the existing flight controller.

DESIGN TARGET

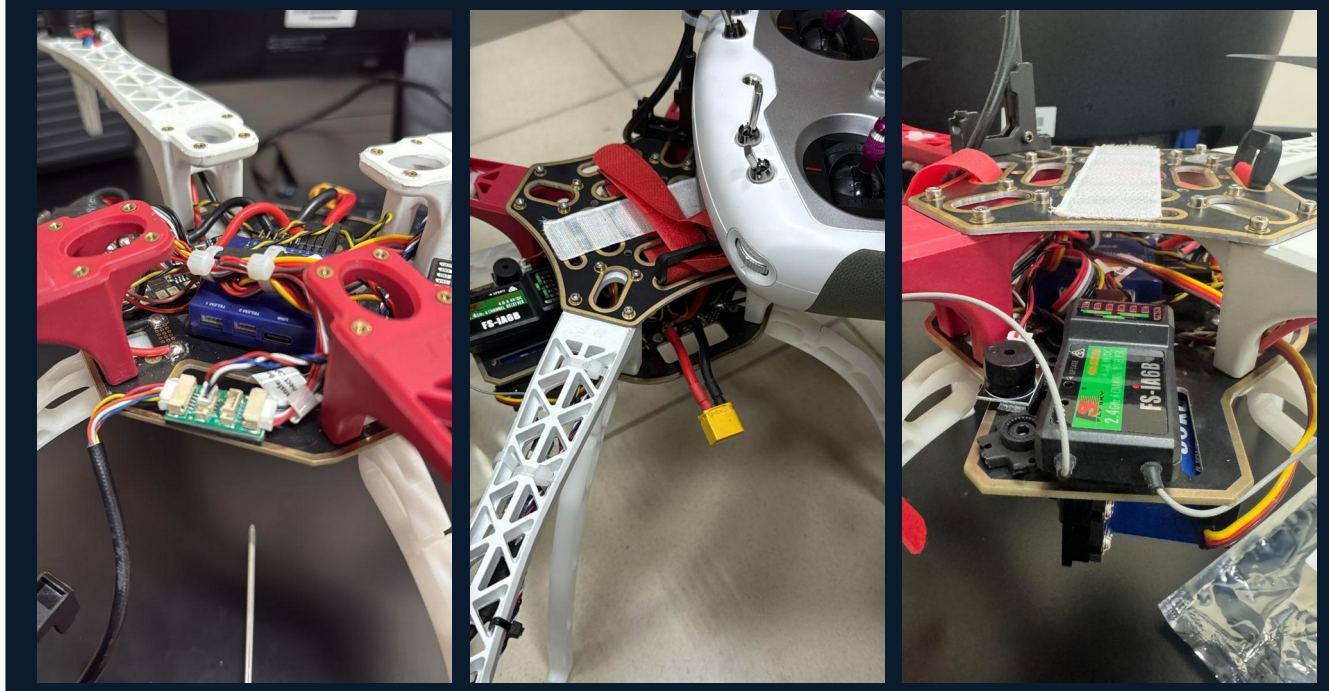
~30 minutes mission endurance

Not yet validated

CURRENT STATUS

The drone is fully assembled and working

Actual assembly photos document the frame, wiring and receiver; AI components are awaiting integration.



Actual assembled drone — airframe, wiring and flight receiver (team photos)

DONE **Drone fully assembled**

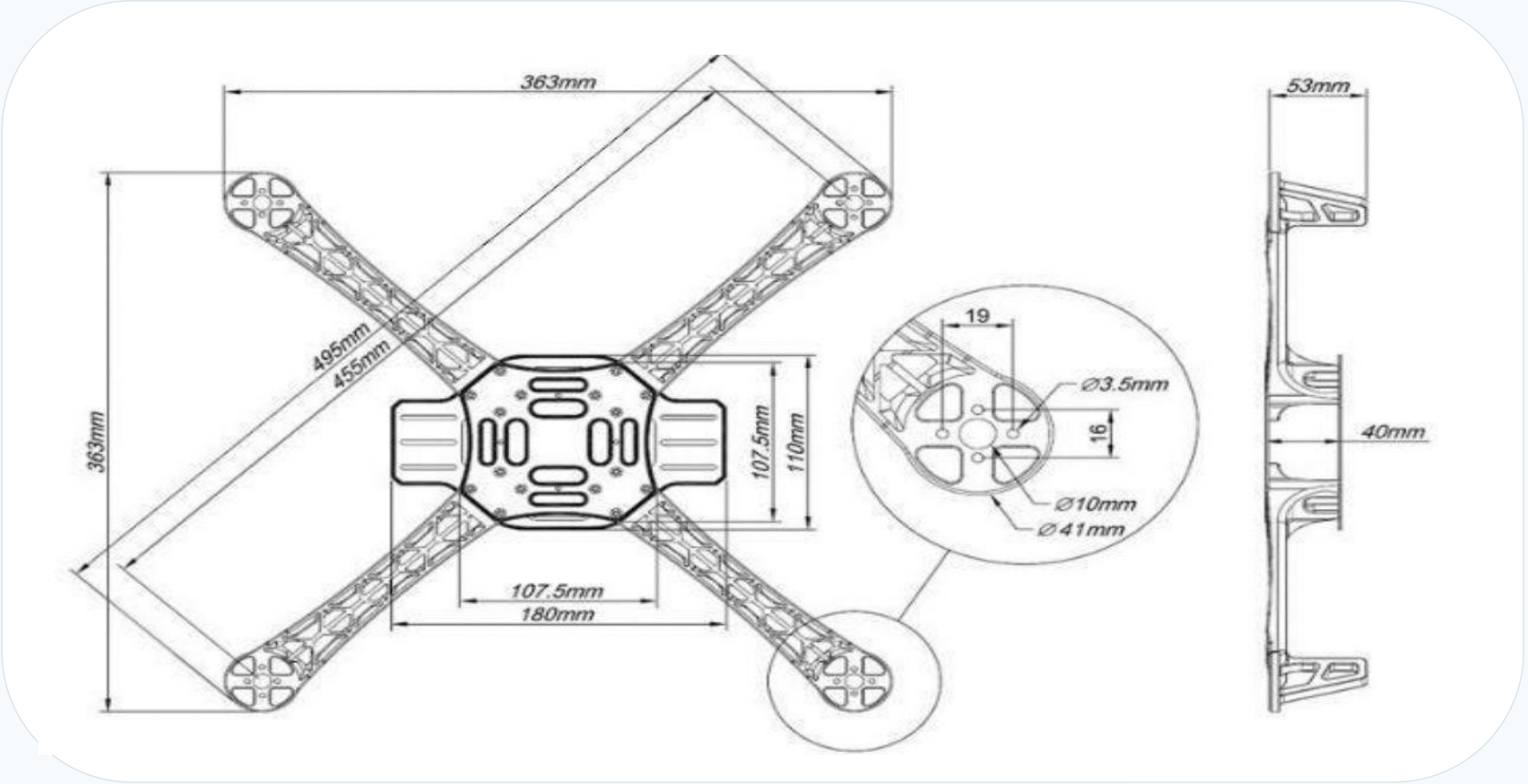
WORKING **Existing flight system**

ORDERED UNO Q processor

Ordered; integration pending

Current quadcopter chassis dimensions

X-frame quadcopter platform replaces the earlier winged VTOL configuration.



DIMENSIONED REFERENCE

363 mm

Span indicated along both drawing axes

180 mm

Center mounting region marked in drawing

Dimensions to be rechecked on actual chassis before manufacturing mounts.

SYSTEM INTEGRATION

Two independent onboard systems

Flight and wildlife detection are separated so the AI does not control the motors.

FLIGHT CONTROL

- Mission Planner waypoint route
- Existing autopilot / stabilization
- ESCs + motors + flight battery

ONBOARD AI DETECTION

- Arduino UNO Q 4GB: ordered
- Camera + YOLO processing pipeline
- Battery-fed AI supply (to verify)

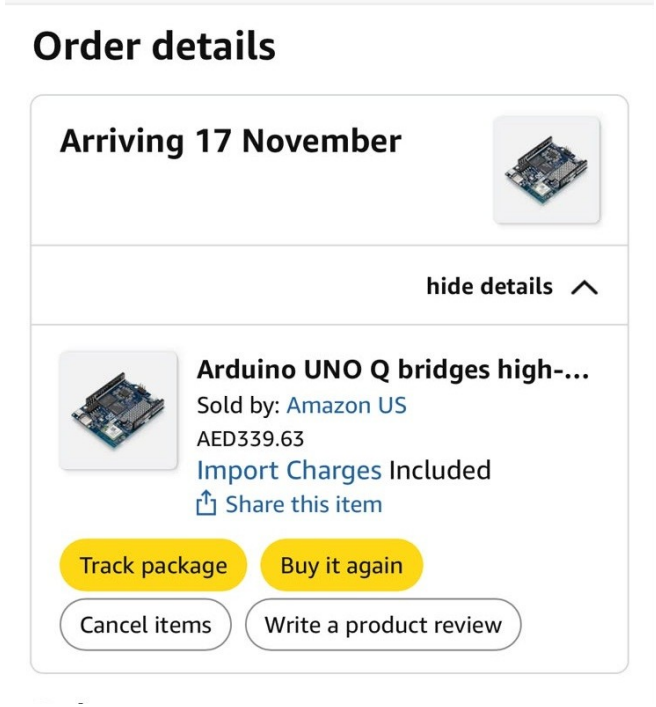
Any connection between systems is for logging / position data, not motor control.

Arduino order and connection accessories

Order confirmed for the UNO Q; power connector options selected for the battery-fed system.

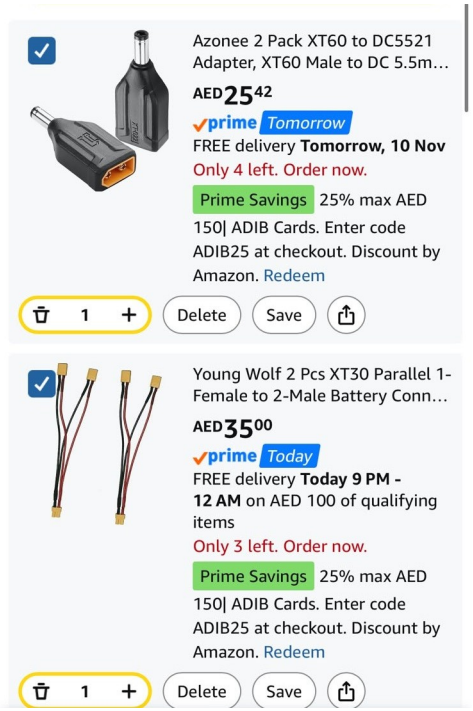
ORDER CONFIRMED

Arduino UNO Q • ETA 17 Nov



IN SHOPPING CART

Battery connector and adapter options



PREPARED HARDWARE INTERFACE

Battery-fed power for onboard AI

A clean connection is prepared; final electrical checks precede Arduino installation.

BATTERY

Existing UAV power source

→

POWER CONNECTION

Secure connector + protected feed

→

ARDUINO UNO Q

YOLO + camera system

INPUT CHECK BEFORE FIRST POWER-UP

UNO Q: 7–24 V at VIN OR regulated 5 V / 3 A via USB-C

Power wiring prepared; not yet demonstrated with the ordered board.

Two blade-manufacturing options explored

Prototypes are planned for fabrication later this week; comparative testing follows.



Toroidal propeller — geometry reference

Reference render; not yet the manufactured blade.

OPTION A

SLS + silicone infusion

Laser-sintered polymer blade with proposed silicone infusion; fabrication planned.

OPTION B

Carbon-fiber 3D printing

Fiber-filled polymer blade; stiffness, mass and surface finish to be evaluated.

2-blade vs 3-blade toroidal geometries | Tests: noise, thrust, RPM and power

EXTERNAL RESEARCH INPUT

Guidance from Ocean Robotics Day

Follow-up with researcher Rawda Alsamahi helped confirm the onboard AI direction.

What we learned from the correspondence

- **Arduino UNO Q 4GB** — Used in her robotic monitoring research.
- **YOLO on-device** — Image preprocessing, inference and postprocessing.
- **Platform selection** — Check physical size, memory, speed and ease of programming.

OUR ADAPTATION

AI on UNO Q

Flight remains on the existing independent autopilot.

Research paper is still under review.

NEXT STEPS

From working drone to validated system

Clear short-term actions and the evidence we will present at the next checkpoint.

1	**Receive components**	Receive UNO Q; finalize camera and verify the battery power connection.
2	**Fabricate blades**	Produce SLS/silicone-infused and carbon-fiber print prototypes later this week.
3	**Bench-test separately**	Run YOLO on the onboard computer; compare thrust, noise and power in a guarded rig.
4	**Integrate & verify**	Install camera/processor; measure mass, power and endurance against targets.

TODAY'S KEY TAKEAWAY

The UAV is operational.
Next: onboard AI + quiet propeller validation.

Instructor feedback requested

Propeller fabrication/testing scope • AI power integration • Endurance target

Thank you

Evidence: current quadcopter photos; chassis dimension drawing; researcher correspondence